

Exercises on Chapter 2: Relations

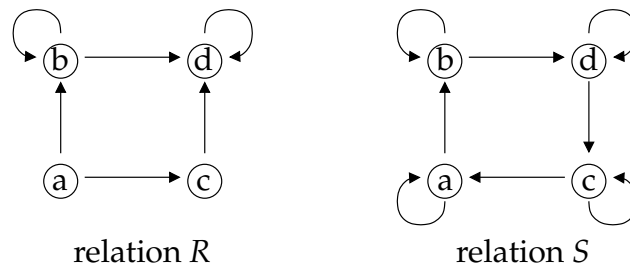
1. In the universe $U := \{a, b, c, d, e\}$ we are given the sets

$$A := \{b, c\} \quad B := \{a, b, c, e\} \quad C := \{a, c, d\} \quad D := \{a, c, e\}$$

- Draw a picture of the sets $B \times C$ and $A \times D$: draw a horizontal and a vertical axis, put the elements of U along these axes, and indicate with the symbol $\times$ which points in the picture are in $B \times C$, and with the symbol $+$ which ones are in $A \times D$ (see slide 4 of the lecture).
- How many elements are there in the set $(B \times C) \cap (A \times D)$, and how many elements are there in the sets $(B \times C) \setminus (A \times D)$ and $B \times C$?
- Use your answers in part (b) to verify that

$$\#((B \times C) \cap (A \times D)) + \#((B \times C) \setminus (A \times D)) = \#(B \times C)$$

- Consider the relation $R := \{\langle 0, 0 \rangle, \langle 0, 1 \rangle, \langle 1, 1 \rangle, \langle 1, 2 \rangle, \langle 2, 1 \rangle\}$ in the set $\{0, 1, 2\}$.
 - Draw a directed graph representation of R .
 - Is the relation R reflexive? transitive? symmetric? anti-symmetric?
 - Explicitly list the elements of R^{-1} in the curly-bracket notation.
- Using the directed graph representation, we define two relations R and S in the set $\{a, b, c, d\}$ as follows:



- Explicitly list the elements of R and S in the curly-bracket notation.
- Give the matrix representations of the relations R and S .
- Determine the relations $R \circ S$ and $S \circ R$ and write down an explicit listing of their elements (in the curly-bracket notation). *Hint: it can help to draw a Venn diagram of the composition first.*
- Draw directed graph representations of the relations $R \circ S$ and $S \circ R$.
- Is the relation $R \circ S$ reflexive? transitive? symmetric? anti-symmetric?
- Is the relation $S \circ R$ reflexive? transitive? symmetric? anti-symmetric?

4. Let R and S be two relations in a set of 4 elements, described by the following matrices (with rows and columns ordered in the same way):

$$R: \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad S: \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Determine the matrix representation of the composite relation $R \circ S$.

Hint: you can use a Venn diagram to picture the composition $R \circ S$ first.

5. In this exercise, we work in a domain of *People*.
- Give an appropriate simplified name to each of the following relations:
 - the relation $IsMarriedTo^{-1}$
 - the relation $(IsBrotherOrSisterOf \circ IsChildOf)^{-1}$
 - the relation $IsParentOf \circ IsBrotherOrSisterOf \circ IsChildOf$
 - Compare the two relations $IsBrotherOf$ and $IsBrotherOf \circ IsBrotherOf$.
 - Is it true that $IsBrotherOf \subseteq (IsBrotherOf \circ IsBrotherOf)$?
 - Is it true that $(IsBrotherOf \circ IsBrotherOf) \subseteq IsBrotherOf$?
 - Express each of the relations below in terms of the relations $IsParentOf$ and $IsTheSamePersonAs$ using the operations of inverse, composition, union, intersection, complement and / or difference:
 - the relation $IsGreatGrandChildOf$
 - the relation $IsBrotherOrSisterOf$
6. In this exercise we describe a date as a triple of the form $\langle d, m, y \rangle$, where $d \in \{01, 02, \dots, 31\}$, $m \in \{01, 02, \dots, 12\}$ and $y \in \{1990, 1991, 1992, \dots\}$. For example, $\langle 13, 04, 1995 \rangle$ denotes the 13th of April, 1995. Let the set *Dates* consist of all triples $\langle d, m, y \rangle$ that correspond to actual dates. We define the relation *Period* in the set *Dates* as the set of all pairs of dates $\langle d_1, d_2 \rangle$ such that d_1 does not lie (strictly) ahead of d_2 , and d_2 lies at most 21 days ahead of d_1 in time.
- Which of the following pairs of dates are elements of *Period*?

$\langle \langle 08, 10, 1998 \rangle, \langle 20, 09, 1998 \rangle \rangle$

$\langle \langle 10, 10, 1998 \rangle, \langle 01, 11, 1998 \rangle \rangle$

$\langle \langle 10, 11, 1998 \rangle, \langle 01, 12, 1998 \rangle \rangle$

- Is the relation *Period* reflexive? transitive? symmetric? anti-symmetric?
- What is the interpretation of the relation $Period \circ Period$?